

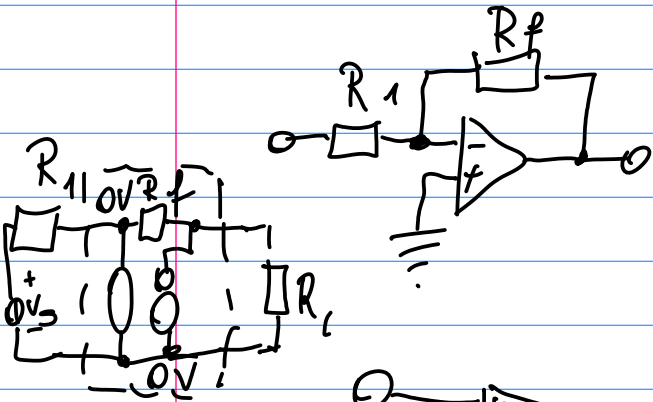
- 1.) Nullor configurations and single rail
- 2.) Biasing error due to high gain and the offset voltage.
- 3.) Common mode range and differential mode range.
- 4.) GB product
- 5.) Signal performance operating conditions
- 6.) Frequency compensation; asymptotic gain model; pole pattern;
- 7.) Resonance of the receive inductor and the parasitics;
- 8.) Receiver is the integrator -
- do the noise study

$$A = \lim_{R_L \rightarrow \infty} \left(\frac{V_o}{V_o} \right)$$

$$B = \lim_{R_L \rightarrow 0} \left(\frac{V_o}{I_o} \right)$$

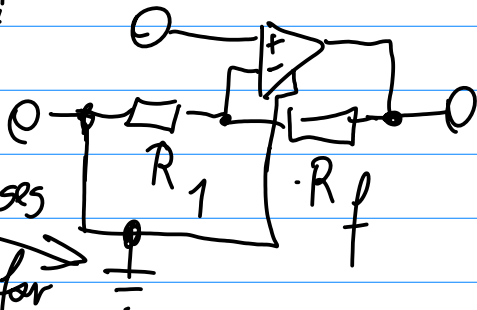
$$C = \lim_{R_L \rightarrow \infty} \left(\frac{I_o}{V_o} \right)$$

$$D = \lim_{R_L \rightarrow 0} \left(\frac{I_o}{I_o} \right)$$



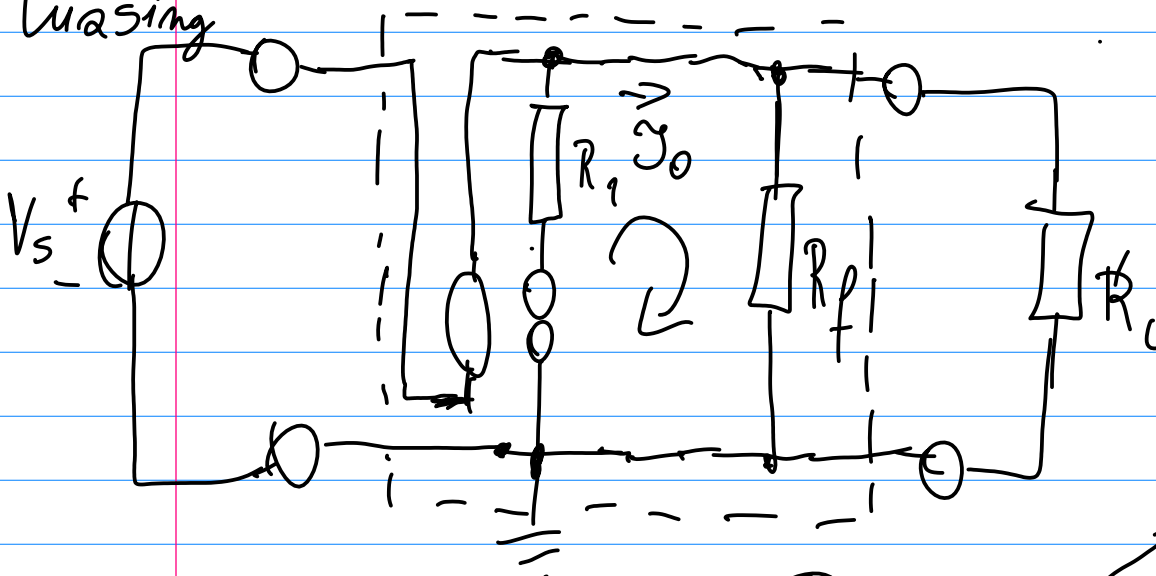
$$\Rightarrow \begin{pmatrix} V_o \\ I_o \end{pmatrix} = \begin{pmatrix} 0 & -R_f \\ 0 & 0 \end{pmatrix} \begin{pmatrix} V_s \\ I_i \end{pmatrix}$$

$$I_i = \frac{V_s}{R_1}$$



$$\begin{pmatrix} V_o \\ I_o \end{pmatrix} = \begin{pmatrix} -\frac{R_f}{R_1} & 0 \\ -\frac{1}{R_1} & 0 \end{pmatrix} \begin{pmatrix} V_s \\ I_i \end{pmatrix}$$

This causes the need for biasing



- ① $I_o R_f = V_o$
- ② $V_i = I_o R_1$

$$\frac{V_o}{R_1} R_f = V_o$$

$$-\frac{R_f}{R_1} = \frac{V_o}{V_o}$$

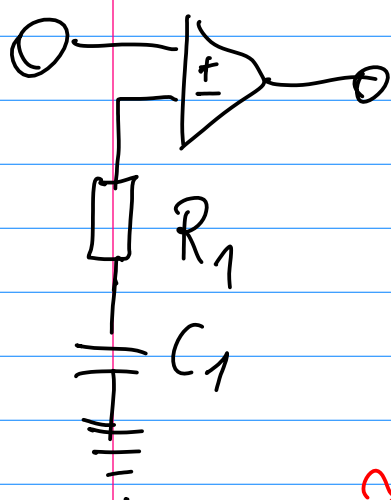
From (2):

$$\frac{I_o}{V_i} = \frac{1}{R_1}$$

$$\begin{pmatrix} V_o \\ I_o \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{L_R} \sqrt{L_S} k_c S}{L_S S + R_S} & \frac{\sqrt{L_R} \sqrt{L_S} k_c S}{L_R S + R_m} \\ \text{stuff} & \frac{\sqrt{L_R} \sqrt{L_S} k_c S}{L_R S + R_m} \end{pmatrix} \begin{pmatrix} V_g \\ I_g \end{pmatrix}$$

For noninverting:

$$\begin{cases} \frac{V_o}{V_g} = -\frac{R_f}{R_1} \\ \frac{I_o}{V_g} = -\frac{1}{R_1} \end{cases} \quad \& \quad R_f = 0 \Rightarrow \begin{cases} \frac{V_o}{V_g} = 0 \\ \frac{I_o}{V_g} = -\frac{1}{R_1} \end{cases}$$



$$\begin{cases} V_{g_{\max}} = 1V \\ I_{o_{\max}} = 320 \text{ mA} \end{cases}$$

From the
pdf attached!

$$\begin{aligned} R_1 &= \frac{1}{0.32} = 3.125 \text{ k}\Omega \approx 3 \text{ k}\Omega \\ Z &= \frac{1}{sC} + R_1 = \frac{1 + sCR_1}{sC} \Rightarrow f = \frac{1}{2\pi RC} \Rightarrow C = \frac{1}{2\pi f R} \end{aligned}$$

The higher the impedance, the lower the I_o , that's make the cap really big and it's transparent.

$$\approx 17 \mu\text{F}$$

Flaring Loop A2.pdf

From the poster:

$$GB_{min} = \frac{2\pi L_s f_{max}^2}{B_T}$$

I trust that the amp from the poster has enough of it. Also for the other op amp!!!

The same goes for input capacitances and output impedances.

After the transformer:

$$k_c = 0.00049$$

$$L_s = 300 \mu H$$

$$L_R = 100 mH$$

$$V_o = I_g \cdot \sqrt{L_R} \sqrt{L_s} k_c S = 0.32 \cdot \sqrt{L_R} \sqrt{L_s} k_c S$$

$$I_o = I_g \cdot \sqrt{L_R} \sqrt{L_s} k_c S / (L_R S + R_n) \quad 8.59 \cdot 10^{-7} S$$

I want to get rid of this non-inverting group does this by being insensitive to I_{in} .

(integrator)

$$\left\{ \begin{array}{l} \frac{V_o}{V_g} = -\frac{R_f}{R_1} \text{ \& } \tau_f = \frac{1}{sC} \Rightarrow V_o = -\frac{1}{sCR_1} V_g \\ \frac{I_o}{V_g} = -\frac{1}{R_1} \end{array} \right.$$

I don't care The receiver detects voltage, not the current.

$$|V_{end}| = \frac{8.59 \cdot 10^{-7}}{C R_1} = 0.25V$$

$$C = 3.9 nF \quad E24$$

$$R = 8.8 k\Omega \quad E96$$

For the biasing:

$$X_1 = \frac{1}{sC} + R_1 \Rightarrow C = \frac{1}{2\pi f} \approx 17 \mu F$$

For the bias network, separation resistor and the DC blocking caps at the input:

$$f = \frac{1}{2\pi RC}$$

$$\Rightarrow 100 k\Omega \& 1 nF$$

